

Event-triggered Control for Networked Control Systems under Denial-of-Service Attack

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Abstract—This paper investigates the networked control systems controller design and synthesis under the denial-of-service (DoS) attacks. The denial of service attack will affect the communication and security performance of the system in the networked control system. In order to ensure the robustness of the system, a relative event-triggered mechanism is proposed, which is applied to the networked control system combined with the observer. Combined with the observer, it is applied to the networked control system, which can alleviate the communication pressure. In this paper, an observer-based control law is designed for the case where the system state is not completely known, the system can also have stable performance under the denial of service attack. Finally, the effectiveness of the proposed method in responding to and eliminating denial of service attacks, is verified by an example of an Internet-based testbed system.

Index Terms—Networked control systems, Stochastic denial-of-service attacks, Resilient observer-based controller

I. INTRODUCTION

With the development of industry and communication technology, networked control systems are applied in many places. For example, power transmission systems, spacecraft, and Unmanned aerial vehicles [1]–[5]. Networked control systems use a network to control devices from a distance. To reduce communication load, they use an event-triggered mechanism. The NCS are susceptible to cyber attacks [6]. Network attacks will affect the confidentiality, integrity and availability of information transmitted by communication channels [7]. For malicious attacks, a controller design lacking security and robustness may lead to system performance degradation or even instability. Unreliable communication networks can further exacerbate these issues, inducing system performance degradation or system instability [8]. Therefore, we should attach great importance to the network security issues in NCSs.

So far, there have been many successes in modeling, analyzing stability, and designing controllers for networked control systems. Network security is crucial in NCSs, including denial of service (DoS) attacks, spoofing attacks, and deception attacks. DoS attacks aim to disrupt transmission and cause packet losses [9]. This paper studies the design of network defense resilience and event-triggered mechanism to enhance NCSs, from the perspective of control theory.

The research topic of Dos attack has been reported, In the control literature. Dos attack model is considered, a state-feedback control strategy [10], aim to mitigate the impact of DoS attacks, which are constrained by frequency and duration, and to provide a sufficient condition to ensure the input-to-state stability of the networked control system. Such as periodic DoS attacks [11], random DoS attacks [12], [13] and arbitrary denial of service attacks [14]–[16]. Effective methods have been developed to analyze and control the behavior of networked control systems under DoS attacks [17]–[19]. In addition, [20] tackle the distributed L2-gain control problem with network system. [21] consider H_2 performance analysis and Distributed under the network.

In order to improve the communication efficiency of the networked control system under the premise of ensuring the control performance. In this paper, an observer-based output feedback controller is designed for discrete-time stochastic networked control systems under denial of service attacks. The main work is as follows, a random distribution is proposed to describe the probability of successful attack, and the probability of dos attack is represented by a random sequence. An event-triggered scheme (ETS) [22] is proposed, which can make the system more resistant to DOS attacks. Progressively, communication resources can be saved and performance levels can be met. Finally, an improved theorem is given so that the nonlinear system can also satisfy this condition.

The contributions of this paper can be summarized as follows:

- The Event-Triggered communication has been proposed for network control systems to enhance communication efficiency and alleviate channel congestion.
- We develop the time delay model to characterize the behavior of the system under DoS attacks, with an opportunity for co-designing the controller gains k and event-triggered parameters δ .

In this paper, we tackle the challenge of designing a controller for Networked Control Systems (NCSs) with an observer in the presence of stochastic Denial of Service (DoS) attacks. We develop a relative event-trigger scheme. Section III introduces new sufficient conditions to ensure the stability and H_∞ performance of the system. In Section IV, we present an example to illustrate the effectiveness of our proposed design

method. Finally, Section V concludes with a summary of our findings.

Notation: Let $\mathbb{R}^{m \times n}$ present the real $m \times n$ matrices and natural numbers. The matrix $\mathcal{Q} \in \mathbb{R}^{n \times n}$ is positive. $\text{He}(\mathcal{G})$ equivalent to $\mathcal{G} + \mathcal{G}^T$ and the notation $*$ in the matrix represents the transpose of matrix by symmetry. Let $\mathcal{L}_2$ denote the space of square-integrable functions defined on the interval $[0, \infty)$.

II. PROBLEM FORMULATION AND PRELIMINARIES

In this following chapters, we first provide system descriptions, design event-triggering rules, and develop controllers based on observers. Then demonstrate that the system remains stable under DoS conditions in Section III.

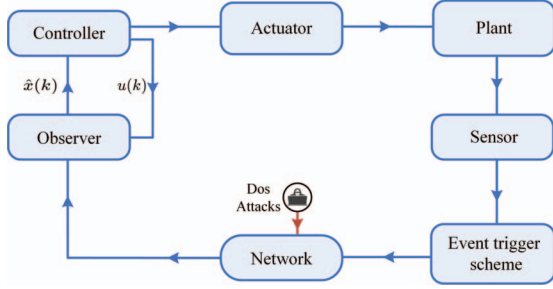


Fig. 1: Network Control System Framework with ETC.

A. System model

Consider the NCSs with stochastic.

$$\begin{cases} x(k+1) = Ax(k) + Bu(k) + Qw(k), \\ y(k) = Cx(k), \end{cases} \quad (1)$$

where $x(k) \in \mathbb{R}^n$ is the system state. The notation $u(k) \in \mathbb{R}^m$ is the control input, and $y(k) \in \mathbb{R}^q$ is the measurement output. $w(k) \in \mathbb{R}^p$ is the system external disturbance, $w(k) \in \mathcal{L}_2[0, \infty)$. The A, B, C and Q are constant matrices, and have appropriate dimensions and their values are given. The symbol $n > q$. The matrices C of row rank is full.

B. Event-Triggered Scheme

The system outputs are sampled periodically. The event-triggered scheme is introduced to determine whether the data is transmitted. If the event trigger communication scheme (ETC) is satisfied, the sensor sampling data will be transmitted to the cache by network, and aim to reduce resource consumption, have better performance indicators. Conversely, data will not be released. The following ETS is defined:

$$e^T(k, k_s) W e(k, k_s) \geq \delta y^T(k) W y(k), \quad k \in (k_s, k_{s+1}), \quad (2)$$

where $e(k, k_s) = y(k_s) - y(k)$, $y(k) = Cx(k)$, $y(k_s) = Cx(k_s)$, $y(k, k_s) = y(k_s) - y(k)$. The constant $\delta > 0$, and weight matrix $W > 0$.

Let $k \in \mathbb{N}$, k denotes current time sampling instant, k_s ($k_s \in \mathbb{N}$) represents the last triggering moment when transmission is successful.

Note that $y(k)$ is denotes current time sampling instant output. If sampling system states $y(k)$ satisfies the ETS (2). Then the system states $y(k)$ is transmitted to the buffer through the network successfully. At this trigger instant, the $y(k)$ is put into $y(k_s)$ by cache, k_s is defined released successfully instant. Alternatively, if not, the sampled data will not be released, then Then wait for the next sampling instant.

Remark 1: Motivated by works [22], We choose relative event-triggered events, similar to it, the data $y(k_s)$ is used in this paper. The released date $y(k_s)$ is more logical and practical with the Stochastic DoS Attacks.

C. Model of the Stochastic DoS Attacks

The data is sampled and transmitted through the network by sensor. In this process, the DoS attacks will prevent data from being transferred to the control center, resulting in delay and packet loss, making data unable to be transmitted.

We describe the possibility of Dos attacks by a random variable $\theta(k_s)$. At every triggering instant, $\theta(k_s) = 0$ implies that the network is attacked, and the data could packet loss, lead some data can not be transmitted. On the other hand $\theta(k_s) = 1$ means the system data released if satisfied the event triggering condition. The data can be transmitted normally over the network without being attacked. Under this condition, the Dos attacks model can be depicted:

$$\tilde{y}(k) = \theta(k_s)y(k_s) + (1 - \theta(k_s))\tilde{y}(k_s - 1), \quad k \in (k_s, k_{s+1}) \quad (3)$$

where $\theta(k)$ obeys a Bernoulli distribution. The probability $P\{\theta(k_s) = 0\} = 1 - \bar{\theta}$ present the DoS attack probability. On the contrary, the symbol $P\{\theta(k_s) = 1\} = \bar{\theta}$ means that the DoS attacks not happened.

In order to conform to logic and practice, we define the initial $\tilde{y}(k_s) = 0$. Then the $\{\beta(k)\}$ is proposed random variable sequence, it defined as:

$$\beta(k) = \theta(k_s), \quad k \in [k_s, k_{s+1}), \quad s \in \mathbb{N}. \quad (4)$$

Base on the definition (3), for $k \in (k_s, k_{s+1})$. Combining (4) with (3), we have

$$\begin{aligned} \tilde{y}(k) &= \beta(k)(y(k) + y(k, k_s)) \\ &\quad + (1 - \beta(k))\tilde{y}(k - 1), \quad k \in [k_s, k_{s+1}), \end{aligned} \quad (5)$$

where $P\{\beta(k) = 1\} = \bar{\theta}$, $P\{\beta(k) = 0\} = 1 - \bar{\theta}$.

Remark 2: Motivated by works [23], which introduced the Bernoulli variable to characterize the effect of network attack. We model the probability of being attacked at each moment, which aligns with physical and real-world logic.

D. Controller Design with Observer

The controller is designed by observer:

$$\begin{cases} \hat{x}(k+1) = (A - LC)\hat{x}(k) + Bu(k) + L\tilde{y}(k), \\ u(k) = K\hat{x}(k), \end{cases} \quad (6)$$

where $\hat{x}(k) \in \mathbb{R}^n$ is the observer states. L and K will be designed gain matrices.

Combining (1), (5) and (6), we have the state equation as follows:

$$\begin{aligned} \epsilon(k+1) = & (A - LC)\epsilon(k) + Qw(k) \\ & - (1 - \theta(k_s))L(\tilde{y}(k-1) - Cx(k)) \\ & - \theta(k_s)LCE(k, k_s), \end{aligned} \quad (7)$$

Where $\epsilon(k) = x(k) - \hat{x}(k)$ and $\xi(k) = [x^T(k) \ \epsilon^T(k) \ \tilde{y}^T(k-1)]^T$,

Then, we have system state as:

$$\begin{cases} \xi(k+1) = [\Pi_1 + (\theta(k_s) - \bar{\theta})\Pi_2]\xi(k) \\ \quad + \theta(k_s)\Pi_3e(k, k_s) + \Pi_4w(k), \\ y(k) = \Pi_5\xi(k), \end{cases} \quad (8)$$

where

$$\begin{aligned} \Pi_1 &= \begin{bmatrix} A - BK & -BK & 0_{n \times q} \\ (1 - \bar{\theta})LC & A - LC & (\bar{\theta} - 1)L \\ \bar{\theta}C & 0_{q \times n} & 0_{q \times n} \end{bmatrix}, \\ \Pi_2 &= \begin{bmatrix} 0_{n \times n} & 0_{n \times n} & 0_{n \times q} \\ -LC & 0_{n \times n} & L \\ -C & 0_{q \times n} & I_q \end{bmatrix}, \\ \Pi_3 &= \begin{bmatrix} 0_{n \times n} \\ -LC \\ C \end{bmatrix}, \quad \Pi_4 = \begin{bmatrix} Q \\ Q \\ 0_{q \times p} \end{bmatrix}, \\ \Pi_5 &= \begin{bmatrix} C^T \\ 0_{n \times q} \\ 0_{q \times q} \end{bmatrix}^T. \end{aligned}$$

Before the next section, two important definitions are used to proof exponentially mean-square stable (EMSS).

Definition 1 [24]: The system (8) is stable with exponentially mean-square (EMSS) under $w(k) = 0_{p \times 1}$, if exists $\xi(0) \in \mathbb{R}^n$, the scalars $a > 0$ and $b \in (0, 1)$ as follow:

$$\mathbb{E}\{\|\xi(k)\|^2\} \leq ab^k \mathbb{E}\{\|\xi(0)\|^2\}. \quad (9)$$

Definition 2 [24]: The system (8) is EMSS with a prescribed H_∞ performance level $\gamma > 0$, if satisfy the conditions (10) as follow:

$$\mathbb{E}\left\{\sum_{k=0}^{\infty} \lambda^k \|y(k)\|^2\right\} < \gamma^2 \mathbb{E}\left\{\sum_{k=0}^{\infty} \|w(k)\|^2\right\} \quad (10)$$

The purpose of this article is to determine the event-triggering parameters δ and to design the controller gain K and observer gain L , while meeting the specified control objectives.

- The system (8) is EMSS with disturbance and stochastic DoS attacks.

- The system (8) have a performance level $\gamma > 0$.

III. MAIN RESULT

This paper introduces two Theorems. Theorem 1 proves the system's stability under DoS attacks. Building upon Theorem 1, Theorem 2 coordinates the design of system parameters, incorporating the design of event-triggering mechanism parameters δ and controller gains K .

Theorem 1: Consider the system (8) under ETS(2), For given λ in the range $(0, 1)$, $\bar{\theta} \in [0, 1]$, and $\delta \in (0, 1]$, the system satisfied the EMSS condition with the H_∞ performance parameters $\gamma > 0$, if there exist a scalar $\rho > 0$. The P is a positive matrix, $P \leq \rho I_{2n+q}$, and $W > 0$ satisfying the following:

$$\Xi = \begin{bmatrix} \Xi_{11} & \Xi_{12} & \Xi_{13} \\ * & \Xi_{22} & \Xi_{23} \\ * & * & \Xi_{33} \end{bmatrix} < 0 \quad (11)$$

where

$$\begin{aligned} \Xi_{11} &= \Pi_1^T P \Pi_1 - \lambda P + \bar{\theta}(1 - \bar{\theta}) \Pi_2^T P \Pi_2 \\ &\quad + \text{diag}\{\delta \bar{W}, 0_{n \times n}, 0_{q \times q}\} + \Pi_5^T P \Pi_5, \\ \Xi_{12} &= \bar{\theta} \Pi_1^T P \Pi_3 - \bar{\theta}(1 - \bar{\theta}) \Pi_2^T P \Pi_3, \\ \Xi_{13} &= \Pi_1^T P \Pi_4, \quad \bar{W} = C^T W C, \\ \Xi_{22} &= \bar{\theta} \Pi_3^T P \Pi_3 - \bar{W}, \quad \Xi_{23} = \bar{\theta} \Pi_3^T P \Pi_4, \\ \Xi_{33} &= \Pi_4^T P \Pi_4 - \gamma^2 I_p. \end{aligned}$$

proof: Choose the candidate formula as:

$$V(k) = \xi^T(k) P \xi(k). \quad (12)$$

Notice that $\mathbb{E}\{\theta(k_s)\} = \bar{\theta}$, $\mathbb{E}\{(\theta(k_s) - \bar{\theta})^2\} = \bar{\theta}(1 - \bar{\theta})$, $\mathbb{E}\{\theta^2(k_s)\} = \bar{\theta}$, and $\mathbb{E}\{\theta(k_s)(\theta(k_s) - \bar{\theta})\} = \bar{\theta}(1 - \bar{\theta})$. Then, can evaluated (12) as

$$\begin{aligned} & \mathbb{E}\{V(k+1) - \lambda V(k) + y^T(k)y(k) - \gamma^2 w^T(k)w(k)\} \\ & \leq \bar{\theta} \xi^T(k) \Pi_1^T P \Pi_3 e(k, k_s) + \bar{\theta} e(k, k_s)^T \Pi_3^T P \Pi_1 \xi(k) \\ & \quad + \bar{\theta}(1 - \bar{\theta}) \xi^T(k) \Pi_2^T P \Pi_3 e(k, k_s) \\ & \quad + \bar{\theta}(1 - \bar{\theta}) \xi^T(k) \Pi_2^T P \Pi_2 \xi(k) \\ & \quad + \bar{\theta} e^T(k, k_s) \Pi_3^T P \Pi_4 w(k) + \bar{\theta} w^T(k) \Pi_4^T P \Pi_3 e(k, k_s) \\ & \quad + \xi^T(k) \Pi_1^T P \Pi_4 w(k) + w^T(k) \Pi_4^T P \Pi_1 \xi(k) \\ & \quad + \xi^T(k) (\Pi_1^T P \Pi_1 - \lambda P + \Pi_5^T P \Pi_5) \xi(k) \\ & \quad + w^T(k) (\Pi_4^T P \Pi_4 - \gamma^2 I_p) w(k) \\ & \quad + \bar{\theta} e^T(k, k_s) \Pi_3^T P \Pi_3 e(k, k_s). \end{aligned} \quad (13)$$

System (8) stays in the ETS (2) as follow:

$$e^T(k, k_s) W e(k, k_s) \leq \delta y^T(k) W y(k) \quad (14)$$

It follows from (13) and (14) that

$$\begin{aligned}
& \mathbb{E}\{V(k+1) - \lambda V(k) + y^T(k)y(k) - \gamma^2 w^T(k)w(k)\} \\
& \leq \bar{\theta} \xi^T(k) \Pi_1^T P \Pi_3 e(k, k_s) + \bar{\theta} e^T(k, k_s)^T \Pi_3^T P \Pi_1 \xi(k) \\
& \quad + \bar{\theta}(1 - \bar{\theta}) \xi^T(k) \Pi_2^T P \Pi_3 e(k, k_s) \\
& \quad + \bar{\theta}(1 - \bar{\theta}) \xi^T(k) \Pi_2^T P \Pi_2 \xi(k) \\
& \quad + \bar{\theta} e^T(k, k_s) \Pi_3^T P \Pi_4 w(k) + \bar{\theta} w^T(k) \Pi_4^T P \Pi_3 e(k, k_s) \\
& \quad + \xi^T(k) \Pi_1^T P \Pi_4 w(k) + w^T(k) \Pi_4^T P \Pi_1 \xi(k) \\
& \quad + \xi^T(k) (\Pi_1^T P \Pi_1 - \lambda P + \Pi_5^T \Pi_5) \xi(k) \\
& \quad + w^T(k) (\Pi_4^T P \Pi_4 - \gamma^2 I_p) w(k) \\
& \quad + \bar{\theta} e^T(k, k_s) \Pi_3^T P \Pi_3 e(k, k_s) \\
& \quad + \delta y^T(k) W y(k) \\
& \quad - e^T(k, k_s) W e(k, k_s).
\end{aligned} \tag{15}$$

which further have

$$\begin{aligned}
& \mathbb{E}\{V(k+1) - \lambda V(k) + y^T(k)y(k) - \gamma^2 w^T(k)w(k)\} \\
& \leq \begin{bmatrix} \xi^T(k) \\ e^T(k, k_s) \\ w^T(k) \end{bmatrix}^T \begin{bmatrix} \Xi_{11} & \Xi_{12} & \Xi_{13} \\ * & \Xi_{22} & \Xi_{23} \\ * & * & \Xi_{33} \end{bmatrix} \begin{bmatrix} \xi(k) \\ e(k, k_s) \\ w(k) \end{bmatrix}, \tag{16}
\end{aligned}$$

when $w(k) = 0_{p \times 1}$, we have

$$\begin{aligned}
& \mathbb{E}\{V(k+1) - \lambda V(k)\} \\
& \leq \begin{bmatrix} \xi^T(k) \\ e^T(k, k_s) \end{bmatrix}^T \begin{bmatrix} \Xi_{11} - \Pi_5^T \Pi_5 & \Xi_{12} \\ * & \Xi_{22} \end{bmatrix} \begin{bmatrix} \xi(k) \\ e(k, k_s) \end{bmatrix}, \tag{17}
\end{aligned}$$

From (17), it follows that

$$\mathbb{E}\{\|\xi(k)\|^2\} \leq \vartheta \lambda^k \mathbb{E}\{\|\xi(0)\|^2\}, \tag{18}$$

where $\vartheta = \frac{\lambda_{\max}\{P\}}{\lambda_{\min}\{P\}}$, system (8) is EMSS by Definition 1 in this condition.

According to (15), for any nonzero vector $w(k)$ we have

$$\mathbb{E}\{y^T(k)y(k) - \gamma^2 w^T(k)w(k)\} \leq \mathbb{E}\{-V(k+1) + \lambda V(k)\}. \tag{19}$$

Summing up iterations of (19), further get

$$\begin{aligned}
& \sum_{k=0}^{\infty} \lambda^k \mathbb{E}\{\|y(k)\|^2\} - \gamma^2 \sum_{k=0}^{\infty} \lambda^k \mathbb{E}\{\|w(k)\|^2\} \\
& \leq \lambda^\infty \mathbb{E}\{\xi^T(0)P\xi(0)\} - \mathbb{E}\{\xi^T(\infty)P\xi(\infty)\}, \tag{20}
\end{aligned}$$

For $\mathbb{E}\{\xi^T(\infty)P\xi(\infty) > 0\}$, as stated in Definition 2. Theorem 1 ensures that system (8) is EMSS and achieves the specified H_∞ performance level. This completes the proof. Due to the challenges posed by the nonlinear term in the design of observer-based controllers, we introduce additional slack matrices to enhance the theorem based on Theorem 1.

Theorem 2: For given $\bar{\theta} \in [0, 1]$, $\delta \in (0, 1]$, $\lambda \in (0, 1)$ and matrix R_1 , the system is EMSS and achieves an H_∞ performance level $\gamma > 0$, if there exist matrices $P = \text{diag}\{P_{11}, P_{22}, P_{33}\} > 0$, $\bar{W} > 0$, $J > 0$, $P \leq \rho I_{2n+q}$, R_2 , Z_1 , and S_1 of appropriate dimensions such that the following inequality holds:

$$\begin{bmatrix} \Phi_{11} & \Phi_{12} \\ * & \Phi_{22} \end{bmatrix} < 0, \tag{21}$$

where

$$\begin{aligned}
\Phi_{11} &= M + \bar{X}_2^T J \bar{X}_2, \\
M &= \begin{bmatrix} M_{11} & M_{12} & M_{13} & M_{14} & M_{15} \\ * & M_{22} & 0_{n \times p} & M_{24} & M_{25} \\ * & * & M_{33} & \Pi_4^T P & M_{35} \\ * & * & * & -P & M_{45} \\ * & * & * & * & -P \end{bmatrix}, \\
M_{11} &= -\lambda P + \text{diag}\{\delta \bar{W}, 0_{n \times n}, 0_{q \times q}\} + \Pi_5^T \Pi_5, \\
M_{13} &= 0_{2n+q \times p}, \quad \phi = \sqrt{\bar{\theta}(1 - \bar{\theta})}, \\
M_{14} &= \begin{bmatrix} M_{14}^{11} & M_{14}^{12} & M_{14}^{13} \\ M_{14}^{21} & M_{14}^{22} & 0_{n \times q} \\ 0_{q \times n} & M_{14}^{32} & M_{14}^{33} \end{bmatrix}, \\
M_{14}^{11} &= A^T P_{11} + S_1^T R_1^T, \quad M_{14}^{12} = (1 - \bar{\theta}) C^T R_2, \\
M_{14}^{13} &= \bar{\theta} C^T P_{33}, \quad M_{14}^{21} = -S_1^T R_1^T, \quad M_{14}^{22} = A^T P_{22} - C^T R_2, \\
M_{14}^{32} &= (\bar{\theta} - 1) R_2, \quad M_{14}^{33} = (1 - \bar{\theta}) P_{33}, \\
M_{15} &= \phi \begin{bmatrix} 0_{n \times n} & M_{15}^{12} & M_{15}^{13} \\ 0_{n \times n} & 0_{n \times n} & 0_{n \times q} \\ 0_{q \times n} & M_{15}^{32} & M_{15}^{33} \end{bmatrix}, \\
M_{15}^{12} &= -C^T R_2, \quad M_{15}^{13} = -C^T P_{33}^T, \\
M_{15}^{32} &= R_2, \quad M_{15}^{33} = -P_{33}, \quad M_{22} = -\bar{W}, \\
M_{24} &= \bar{\theta} [0_{n \times n} - C^T R_2 C^T P_{33}], \\
M_{25} &= \phi [0_{n \times n} - C^T R_2 C^T P_{33}], \\
M_{33} &= -\gamma^2 I_p, \quad M_{35} = 0_{p \times (2n+q)}, \quad M_{45} = 0_{(2n+q) \times (2n+q)}, \\
\Phi_{21} &= [X_1^T \quad 0_{(7n+3q+p) \times n}], \\
X_1 &= [R_1^T R_1 S_1 \quad -R_1^T R_1 S_1 \quad 0_{m \times (5n+3q+p)}], \\
\bar{X}_2 &= [0_{n \times (3n+q+p)} \quad I_n \quad 0_{n \times (3n+2q)}], \\
\Phi_{22} &= \begin{bmatrix} \text{He}\{-R_1^T R_1 Z_1\} & (P_{11} B - R_1 Z_1)^T \\ * & -J \end{bmatrix}.
\end{aligned}$$

Then the closed-loop system (8) is EMSS with with a prescribed H_∞ performance level and $\gamma > 0$, controller and observer gain and the weight matrix as

$$\begin{aligned}
K &= Z_1^{-1} S_1, \quad L = P_{22}^{-1} R_2^T, \\
W &= (C C^T)^{-1} C \bar{W} C^T (C C^T)^{-1}.
\end{aligned}$$

proof: By Schur complement and inequality (11), we can further get

$$\begin{bmatrix} M_{11} & 0_{n \times n} & M_{13} & \Pi_1^T P & \pi \Pi_2^T P \\ * & M_{22} & 0_{n \times p} & \bar{\theta} \Pi_3^T P & \phi \Pi_3^T P \\ * & * & M_{33} & \Pi_4^T P & M_{35} \\ * & * & * & -P & M_{45} \\ * & * & * & * & -P \end{bmatrix} < 0. \tag{22}$$

From (22), we have

$$\begin{aligned} \bar{M} + \bar{X}_1 \bar{X}_2 < 0, \\ \bar{\Sigma} = \begin{bmatrix} M_{11} & 0_{n \times n} & M_{13} & \bar{M}_{14} & \bar{M}_{15} \\ * & M_{22} & 0_{n \times p} & \bar{M}_{24} & \bar{M}_{25} \\ * & * & M_{33} & \Pi_4^T P & M_{35} \\ * & * & * & -P & M_{45} \\ * & * & * & * & -P \end{bmatrix}, \\ \bar{M}_{14} = \begin{bmatrix} A^T P_{11} & (1 - \bar{\theta}) C^T L^T P_{22} & \bar{\theta} C^T P_{33} \\ 0_{n \times n} & A^T P_{22} - C^T L^T P_{22} & 0_{n \times q} \\ 0_{q \times n} & (\bar{\theta} - 1) L^T P_{22} & (1 - \bar{\theta}) P_{33} \end{bmatrix}, \\ \bar{M}_{15} = \pi \begin{bmatrix} 0_{n \times n} & -C^T L^T P_{22} & -C^T P_{33}^T \\ 0_{n \times n} & 0_{n \times n} & 0_{n \times q} \\ 0_{q \times n} & L^T P_{22} & -P_{33} \end{bmatrix}, \\ \bar{M}_{24} = \bar{\theta} [0_{n \times n} - C^T L^T P_{22} C^T P_{33}], \\ \bar{M}_{25} = \phi [0_{n \times n} - C^T L^T P_{22} C^T P_{33}], \\ \bar{X}_1 = [P_{11} B K \quad -P_{11} B K \quad 0_{m \times (5n+3q+p)}], \end{aligned}$$

Hold if (21) satisfied, we have

$$\bar{M} + \bar{X}_1 J^{-1} \bar{X}_1^T + \bar{X}_2^T J \bar{X}_2 < 0. \quad (23)$$

According to (23), if $P_{11}B = R_1 Z_1$. Therefore, it can be deduced that Equation (21) and Equation (11) are equivalent by Schur's theorem. Then the proof is completed.

Remark 3: Motivated by works [23], we introducing an auxiliary matrix R1 under the relative ETS. To achieve decoupling of the controller gain and observer gain, [25] emphasizes the necessity of ensuring that matrix B has full column rank.

Remark 4: According to Theorem 2, we decouple Theorem 1 to enable the co-design of controller gain k , event-triggering parameters, and the H_∞ performance index in Theorem 2. The size of the event-triggering threshold parameter affects the event-triggering interval. Balancing the event-triggering parameters with system performance will be the focus of future work.

IV. ILLUSTRATIVE EXAMPLE

This section utilizes an Internet-based test rig [26] to demonstrate the proposed design approach with the following parameters:

$$\begin{aligned} A = \begin{bmatrix} 1.12 & 0.213 & -0.333 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}, B = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \\ C = \begin{bmatrix} 0.0541 \\ 0.0050 \\ 0.0001 \end{bmatrix}^T, Q = \begin{bmatrix} 0.2 \\ 0 \\ 0.1 \end{bmatrix}. \end{aligned}$$

Q is the other system parameters, we set $\bar{\theta} = 0.95$, $\lambda = 0.95$, $\delta = 0.5$, $\gamma = 0.2$, by solving LMI in (21), we have

$$\begin{aligned} K = [-0.9424 \quad -0.2149 \quad 0.3282], \\ L = [14.6485 \quad 13.9023 \quad 13.1864]^T, W = 6.4955. \end{aligned}$$

We initialize the system with $x(0) = [0.5 \quad 0.8 \quad 0.6]^T$ and apply an external disturbance modeled as $w(k) =$

$\cos(0.8k)e^{-0.1k}$. Fig.2 and Fig.3 illustrate the state and observer responses, respectively. Figure 5 reveals that data packets are transmitted by the ETC as per condition (2) and experience blockage while traversing the network. Subsequently, we analyze the effects of the event-triggered parameter δ and the probability of no attack, denoted by $\bar{\theta}$. This discussion sheds light on how variations in δ and $\bar{\theta}$ influence the system's behavior and its ability to withstand DoS attacks.

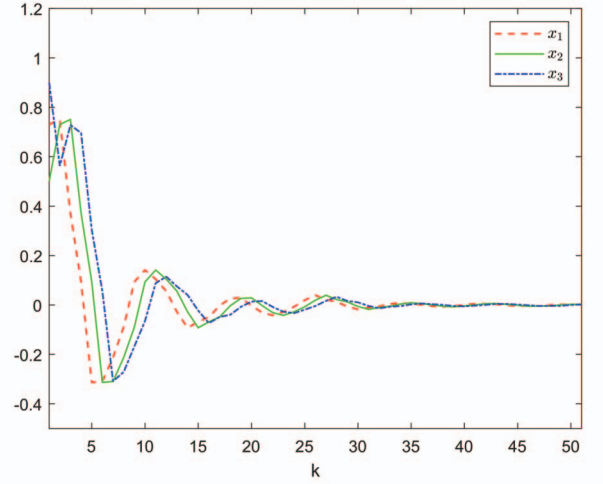


Fig. 2: The state responses

V. CONCLUSIONS

This paper discusses a resilient control approach for networked systems facing external disturbances and DoS attacks. It includes an observer, a relative event-triggered mechanism, and an observer-based controller. DoS attacks on the network between the controlled object and the controller can disrupt normal signal transmission. The proposed relative event-triggered mechanism helps protect against random DoS attacks and saves communication resources by adjusting δ . An event-triggered H_∞ controller has been designed to ensure the stability and performance of NCS in the presence of disturbances and random DoS attacks. This is achieved through a co-design methodology. The effectiveness of the theoretical approach is demonstrated using a DC servo system. The findings show that the proposed strategy significantly mitigates the risk and impact of DoS attacks, resulting in a robust and dependable networked control system.

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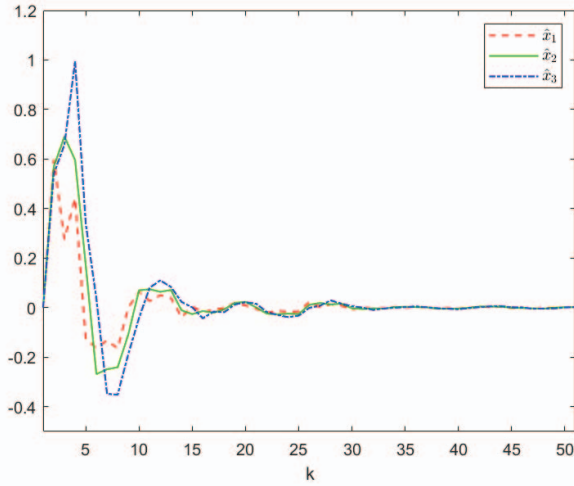


Fig. 3: The observer responses.

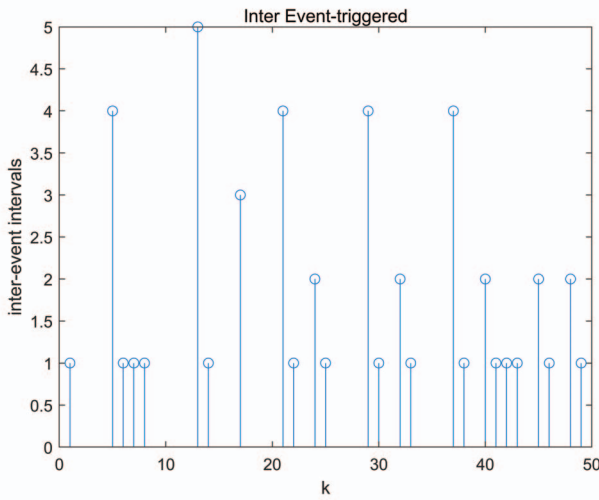


Fig. 4: Inter-event intervals under ETC.

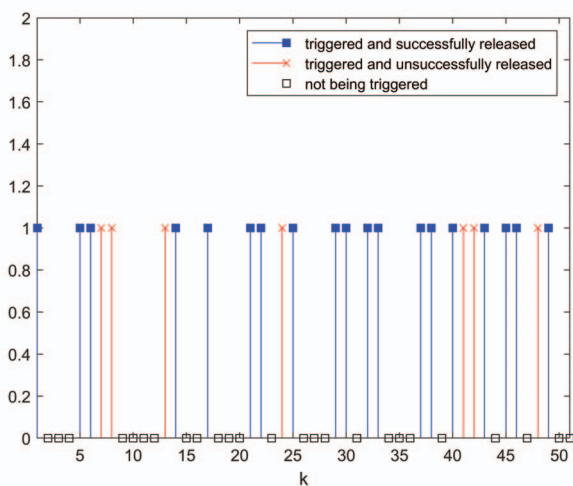


Fig. 5: Three situations of data transmission.

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